The least squares line of best fit

1 4 knots

2

a 8500 m **b** 6500 m **c** 300 seconds **d** 1800 seconds

3

a $20^{\circ}\mathrm{C}$

b 210 Chirps per minute

c 130 Chirps per minute

The equation of a least squares line of best fit

4 $y = 4x - 15$

5

a Negative

b **C**

6

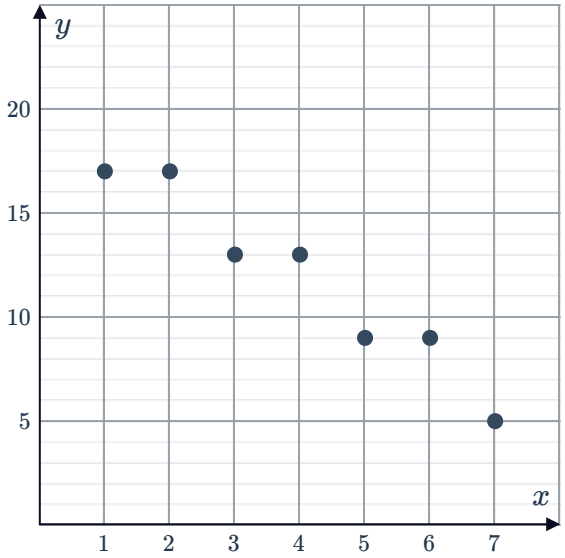
a 57.622 m

b $h = 0.8$

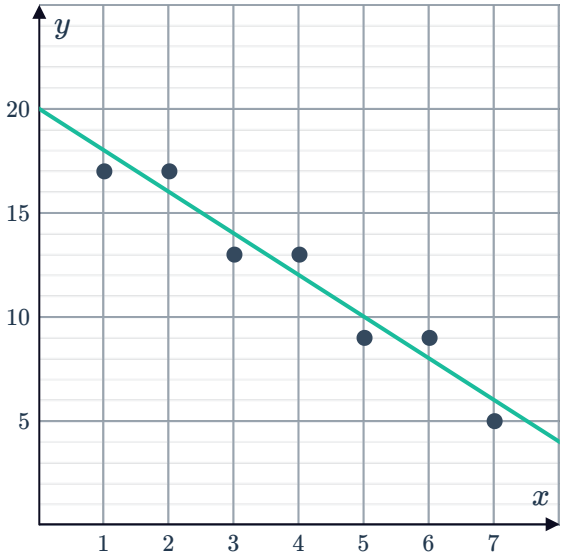
7 Decreasing

8

a



b

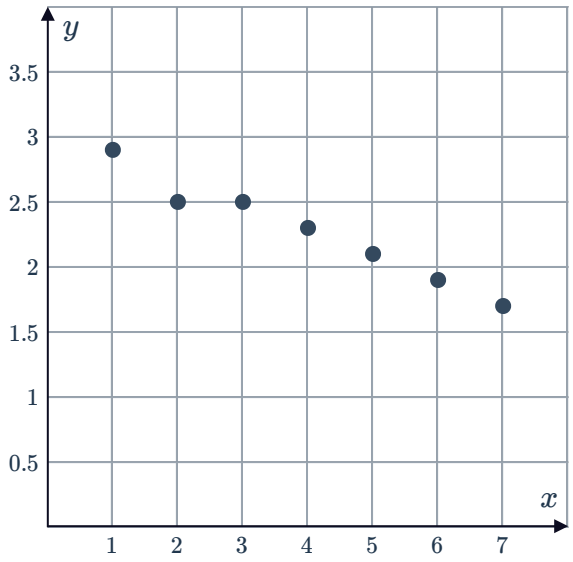


c 0

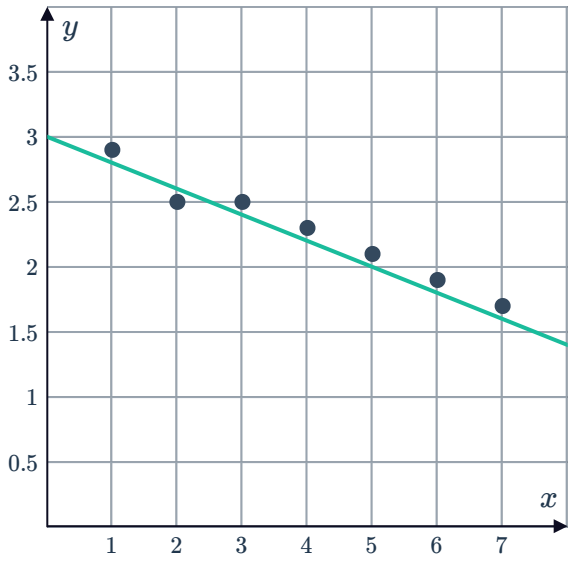
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9

a



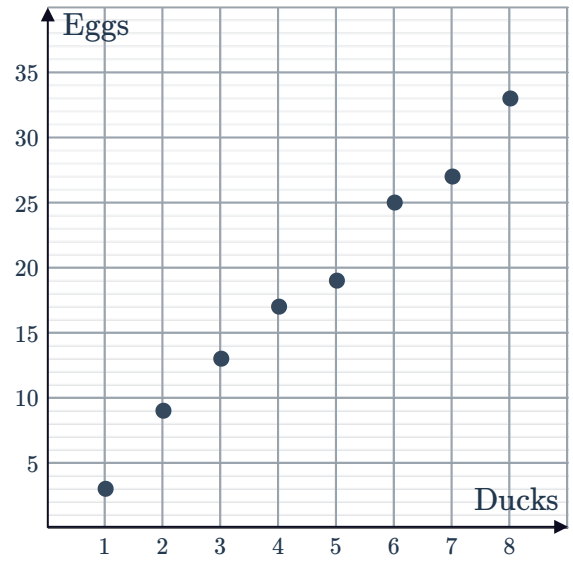
b



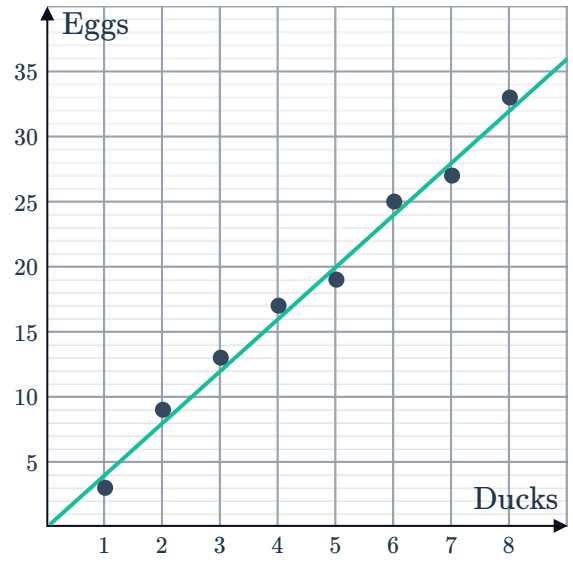
c 0.6 kg of litter collected

10

a



b Example answer:



c 4

d 0

e $y = 4x$

f 96

11

a Negative gradient

b -2.5

c $y = -2.5x + 105$

d $y = 95$

12

a $x = 20, y = 60$

b -3

c $y = -3x + 60$

d 24 hours

13

a 0.5

b $y = 0.5x + 4$

c 7.45

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14

a -1.2 **b** 16 **c** $y = -1.2x + 16$

15

a $y = -0.005x + 49$ **b** $x = 3682$ km

16

a $\frac{1}{16}$ **b** $b = 100$ **c** $\$400\,000$

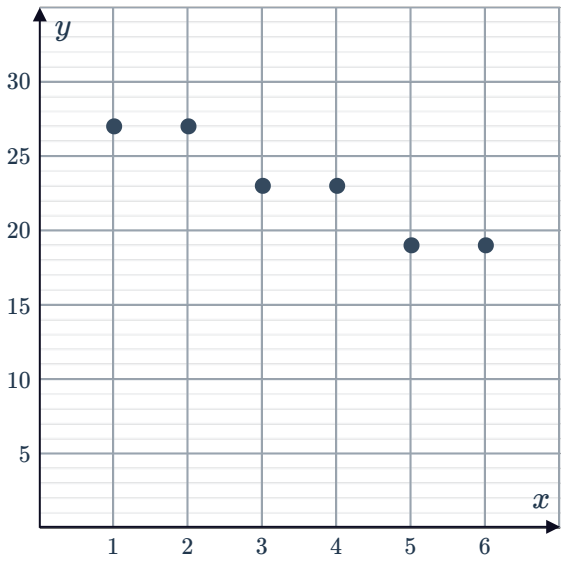
17

a 42 m **b** 2 **c** $b = 30$ **d** 39 m

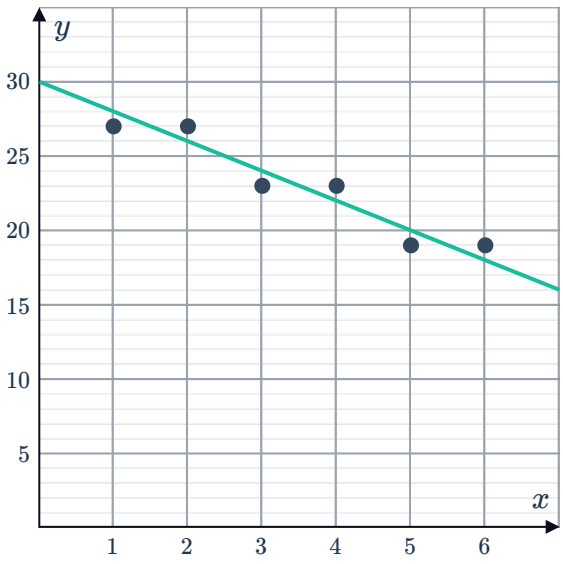
18 $b = -5.6$

19

a



b



c -2 **d** 30

e $y = -2x + 30$ **f** 7.5

20

a $(1040, 54)$ **b** $\frac{1}{40}$ **c** $b = 28$ **d** $L = \frac{1}{40}A + 28$

e 0.675 years

21

a -25 **b** $b = 750$

c $A = -25T + 750$ **d** $T = 30^\circ\text{C}$

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Determining the least squares line using technology

22

a

i -0.85

ii Strong negative linear relationship

iii $y = -0.1x + 30.3$

b

i 0.75

ii Moderate positive linear relationship

iii $y = 5.8x - 99.0$

23

a $m = 3.3$

b $r = 0.95$

c Strong positive linear relationship.

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